

MUJUNI SIMON PETER – SOFTWARE ENGINEER

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PROFILE

I am a results-driven, versatile and proactive team player, whose attention to detail and eagerness to bring new ideas enhance productivity and drive progress in an organization. I have a strong interest in Artificial Intelligence, Machine Learning, computational systems and AI research for African societal challenges. Experienced in building Python-based AI applications, data-driven systems, REST APIs and full-stack software.

My independent project work includes healthcare information systems, emotion recognition, machine-learning-based behavioural modelling, AI safety, cultural and language preservation, and intelligent financial systems.

EDUCATION

Bachelor of Science in Software Engineering (Makerere University)

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, TypeScript, SQL

AI / Machine Learning: Computer Vision, Natural Language Processing, Affective Computing, Predictive Modelling, Data Preprocessing

Python Libraries: NumPy, Pandas, Scikit-learn, OpenCV, PyTorch

Backend: FastAPI, REST APIs, SQLAlchemy, authentication and authorization

Frontend: React, Next.js, HTML, CSS

Databases: PostgreSQL, SQLite, Firebase/Firestore

Tools: Git, GitHub, Vercel, Render, Netlify

NOTABLE PROJECTS

1. Mizizi: AI-Powered Cultural & Language Preservation

Python | AI/NLP | Speech & Language Technology

Developed an AI-oriented platform for preserving Ugandan oral cultural heritage, including stories, songs, riddles, poems and other forms of oral knowledge.

The project explores the use of AI to capture, preserve, organise and make culturally significant knowledge accessible to future generations.

Research relevance:

- NLP for low-resource languages
- Speech and language technology
- Local-language datasets
- Cultural knowledge representation
- AI for African societal applications

GitHub: <https://github.com/sp-mujuni/mizizi>

2. NeuroSim: Healthcare Information & Intelligent Records System

Python | FastAPI | React | SQLAlchemy | PostgreSQL/SQLite

Developed a healthcare information system designed to structure patient information and clinical workflows across registration, appointments, triage, consultation, laboratory services, prescriptions and billing.

Key work:

- Designed a Python/FastAPI backend and web-based interface.
- Implemented structured patient and clinical records.
- Developed authentication and role-based access control.
- Built REST APIs and database models.
- Implemented laboratory, prescription, appointment and billing workflows.
- Explored intelligent processing of structured healthcare information.

Research relevance: Healthcare AI, clinical decision support, structured medical data and intelligent health information systems.

GitHub: <https://github.com/sp-mujuni/NeuroSim>

3. MoodMind: Human Emotion Analysis

Python | Computer Vision | DeepFace | Gemini AI | Firestore

Developed an AI application for analysing human emotions from camera input and generating contextual recommendations.

The project combines computer vision, facial-expression analysis and generative AI to explore how computational systems can interpret and respond to human affect.

Research relevance:

- Computer Vision
- Mental Health
- Affective Computing
- Human-AI Interaction
- Emotion Recognition
- AI-assisted recommendation systems

GitHub: <https://github.com/sp-mujuni/moodmind-app>

4. Zero-Burn: Machine Learning for Developer Burnout

Python | Machine Learning | Data Analysis

Developed a machine-learning-oriented system for analysing factors associated with software-developer burnout and identifying patterns that may indicate increased risk.

Research relevance:

- Predictive modelling
- Data analysis
- Behavioural modelling
- Human-centred AI
- Applied machine learning

GitHub: <https://github.com/sp-mujuni/Zero-Burn-algorithm>

5. Hz: Computational Music Emotion Analysis

Python | AI / Machine Learning

Developed a project exploring computational analysis of emotions expressed through music.

The project forms part of my broader interest in computational modelling of human emotional states.

Research relevance:

- Affective computing
- AI and emotion modelling
- Computational analysis
- Human-centred AI

GitHub: <https://github.com/sp-mujuni/Hz>

6. MindWatch: AI Safety & Psychological Harm Dataset Concept

Python | NLP | AI Safety

Developed a concept for collecting real-world reports of psychological harms associated with AI systems and transforming those narratives into structured material that can support AI-safety evaluation.

Research relevance:

- Natural Language Processing
- Dataset development
- Responsible AI
- AI safety
- Human-AI interaction

GitHub: <https://github.com/sp-mujuni/mindwatch>